



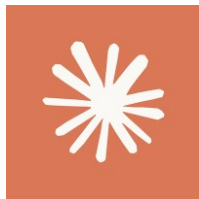
R&D for (Open) Large Language Models in Norway

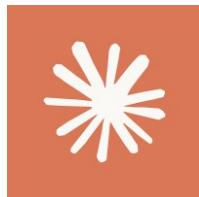
Stephan Oepen, Language Technology Group (LTG), University of Oslo

September 29, 2026 – NorwAI Innovate at Trondheim Tech Festival

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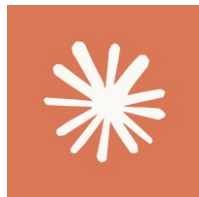
**Background,
Preliminaries**





Cultural Norms and Linguistic Identity

- ▶ *I helgen var **jeg** på hytten i **skogen** og danset med min **søster**.*
- ▶ *I helga var **jeg** på hytta i **skauen** og dansa med **søster**a mi.*
- ▶ *I helga var **eg** på hytta i **skogen** og dansa med **syster**a mi.*
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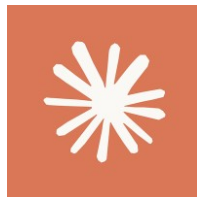
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By Way of Self-Introduction (Gemma-4-21b-it)

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What makes Stephan Oepen at the University of Oslo tick in terms of his academic work?



Reasoning >

Based on his publications, research projects, and academic trajectory, Stephan Oepen's work at the University of Oslo is driven by a desire to bridge the gap between **formal linguistic theory** and **modern machine learning**.

While many in the current NLP (Natural Language Processing) field treat language as a sequence of tokens to be predicted by a "black box" model, Oepen is motivated by the underlying **structure** of language.

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Summary: The "Philosophical" Driver

If you had to distill what drives him, it is the belief that **language is not just a statistical pattern, but a structured system**.

“Hallucinations” Remain a Challenge (September 2026)

Here is a breakdown of what "makes him tick" academically:

1. The Quest for Structure (Parsing)

At the core of Oepen's work is **parsing**—the process of analyzing a string of symbols to determine its grammatical structure. He has spent a significant portion of his career exploring:

- **Dependency Parsing:** How words relate to one another in a hierarchy.

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3. Multilingualism and Low-Resource Languages

A significant motivator for Oepen is the democratization of NLP. Much of the field's progress is centered on English. Oepen focuses on:

- **Cross-lingual Transfer:** Using knowledge from a high-resource language (like English) to help a model understand a low-resource language.
- **Universal Dependencies:** He contributes to the effort of creating a consistent framework for grammar across different languages, allowing for a more unified theory of computational linguistics.

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The ChatGPT Lawyer Explains Himself

In a cringe-inducing court hearing, a lawyer who relied on A.I. to craft a motion full of made-up case law said he “did not comprehend” that the chat bot could lead him astray.



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The Situation in Norway

Februar 5: Renewed and Intensified Collaboration



 Nasjonalbiblioteket

 NTNU



UNIVERSITETET
I OSLO

 sigma2

Språkrådet 

The Norwegian LLM Ecosystem Today (As we See it)

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National Library – «Språkmodellfabrikken»

DEVELOPMENT AV APPLICATION-READY MODELS FOR NORWEGIAN AND SÁMI;
NATIONAL BUDGET 2025 & 2026: BASE FUNDING & RIGHTS, NOT RESEARCH.

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Sigma2 – GPU Capacity and HPC Competencies

LARGE-SCALE E-INFRASTRUCTURES AS ESSENTIAL LLM PRECONDITION;
IN COMPARISON TO E.G. FINLAND OR SWITZERLAND, NORWAY IS BEHIND.

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AI Norge & *New Nordics AI* – AI-Driven Innovation, Regulation, Compliance

BROAD VIEW ON AI; NOT FOCUSSED ON RESEARCH OR LANGUAGE MODELS.

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NTNU & UiO (& National Library) – Foundational and Applied Research

SOME ACTIVITY IN NATIONAL AND EUROPEAN RESEARCH PROJECTS;
ISOLATED AND FRAGMENTED; NO AI CENTER HAS LLMS AS ITS FOCUS.



Språkmodellklynge Norge

Om oss



Nasjonalbiblioteket



NTNU



UNIVERSITETET
I OSLO



sigma2 Språkrådet

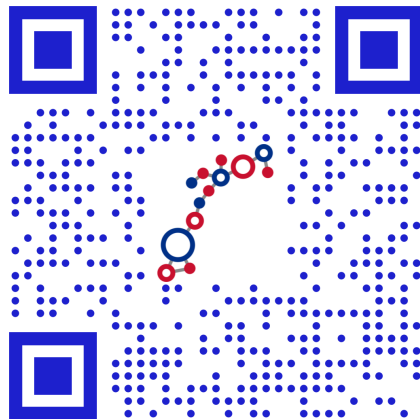
Nasjonal statusmelding: Forskning og utvikling for store, åpne språkmodeller i Norge

kontakt@språkmodellklynge.no

Anbefalinger

Av hensyn til så vel digital som kulturell suverenitet må Norge etablere og drifte et helhetlig nasjonalt økosystem for utvikling, evaluering og anvendelse av egne, trygge og åpne språkmodeller. Det forutsetter blant annet styrket forskningskompetanse og -kapasitet, storskala GPU-infrastruktur, tjenester for data- og modelldeling, samt rammer for samspill mellom grunnleggende forskning og praktisk anvendelse, inkludert styrket nordisk og europeisk samarbeid. Det kreves rask og koordinert handling på regjeringsnivå, inkludert samhandling med for eksempel Norges forskningsråd og Nordisk ministerråd.

Språkmodellklynge Norge anbefaler å:



<https://språkmodellklynge.no/>

1. Initiate a fast-moving process towards a holistic national LLM strategy, with genuine involvement of key Norwegian research environments, together with other national LLM stakeholder.

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2. Create an AI research center for open language models, where all actors come together in an academic partnership, including an associated doctoral training network. Over five years, 15–20 doctoral degrees should be completed.

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3. Further develop Sigma2 as the national provider of compute capacity, LLM services, and HPC expertise, including increased Norwegian commitment to European initiatives like LUMI-AIF and EuroHPC.

4.

Stimulate Nordic–Baltic research and development collaboration via, for example, data and model sharing, researcher mobility, coordinated training, and an annual summit for researchers and developers of open language models.

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Provide the **legal and practical environment** for data sharing, balancing copyright and privacy with the R&D requirements for **large-scale experimentation** and for **inspection and validation** of models.

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5. Provide the legal and practical environment for data sharing, balancing copyright and privacy with the R&D requirements for large-scale experimentation and for inspection and validation of models.
6. Establish Norwegian membership in the European Alliance for Language Technologies (ALT-EDIC) and a financial framework for equitable university participation in the Digital Europe programme.

Moving on: More Cluster Activity in 2026

Ongoing: Invitations to Additional R&D and Strategic Partners

SÁMI RESEARCH GROUPS AT SÁMI ALLASKUVLA AND UIT;
RELATED ACTIVITIES AT UIB, UIS, NR, SIMULA, . . . ; AI NORWAY.

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Shared Research Infrastructure: Data, Evaluation, Software

UNIFORM SETUP ON OLIVIA OG LUMI; STANDARDIZED EVALUATION;
JOINT WORK WITH ALT-EDIC: NORWEGIAN COMPAR:IA LLM ARENA.

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November 30 & December 1: *Nordic–Baltic LLM Builders Summit*

40–50 RESEARCHERS FROM DENMARK, FINLAND, ICELAND, SWEDEN, BALTICS;
«HANDS-ON» COORDINATION; E.G. DATA SHARING, TRAINING, EVALUATION.

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